

Data Engineering and Machine Learning Pipeline

Project Report

ENGE707 Data Engineering and AI

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1. Problem Definition and Dataset Selection

1.1 Problem Statement

There are countless factors in people's daily lives that can influence their health, health decisions such as sleep, physical activity, diet, alcohol intake and other lifestyle habits. Being able to understand how these factors could influence serious health conditions could help people make better lifestyle choices in regards to their health. Heart disease is the biggest global health problem, causing 19.8 million deaths in 2022 alone. Being able to identify lifestyle factors that are associated with heart disease could help people make more cautious health choices, before heart disease develops.

Currently, identification of heart disease requires a major cost as in New Zealand, it approximately cost 538.8 million NZD (Telfar Barnard, 2025). Unaffordability is one of the main causes as well as geographically, and the ability to receive frequent testing for the general population. Even though the primary affected group is the general population, the health care facilities and personnel are also affected. This is because if the government would like to encourage more people to get regular health checks, this would overwhelm the facility thus causing less possibility for individuals that actually need testing. Figuring out the lifestyle that tends to give a higher chance of heart disease by the use of a model would increase its availability to determine whether further actions are needed.

The problem investigated in this project is whether the information of patients such as vitals and lifestyle choices could be used to predict whether or not they have heart disease. Instead of one characteristic, this project will investigate how multiple patient characteristics relate to heart disease and determine which of these variables are useful for accurately predicting heart disease.

The analytical goal of this project is to develop a machine learning model that can predict whether a person belongs to the heart disease category using relevant patient information from data analysis. This would benefit the stakeholders of our project, these being the **general population, healthcare workers, and researchers.**

1.2 Dataset Selection

The dataset we selected for this project is the “*281K Patient Dataset for Multi-Disease Prediction*” sourced from Kaggle on 18 August 2026 and is released under the CCO Public Domain licence.

<https://www.kaggle.com/datasets/danishjmeo/281k-patient-dataset-for-multi-disease-prediction?resource=download>

This dataset contains 280,985 patient records and 39 columns, which include clinically recorded vitals, lifestyle choices, demographic information, and documented diseases. This makes the dataset suitable for our problem statement as it contains all these measurements which can be analysed to form a possible prediction of potential heart disease.

This integrated dataset is a combination of 3 Kaggle datasets that are available publicly. The 3 datasets cover diabetes, heart disease, and hypertension. This source credits the 3 datasets mentioned

(<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>,

[kaggle.com/datasets/oktayrdeki/heart-disease](https://www.kaggle.com/datasets/oktayrdeki/heart-disease),

[kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset](https://www.kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset))

2. Data Acquisition, Inspection, and Documentation

2.1 Data Acquisition and Dataset Structure

The dataset was loaded with the Pandas library in python, and we saved a copy of the dataset before any cleansing or transforming was done. During the initial inspection, python returned **280985 records** and **39 columns**. Python also showed that the data had 3 data types: 19 float columns, 1 int column, and 19 string columns.

Each of the columns describe a different information on each patient, however the main ones can be categorised into demographic, lifestyle, clinical measurements, and disease related information.

Variable Type	Features and Meanings
Demographic	• age, Patient age in years • age_level, Age level is the age group • age_normalized, Normalized age (0,1) • gender, Gender • education_level, Education level • employment_status, Employment status
Lifestyle	• smoking, Whether the instance smokes or not • sleep_hours, Average daily sleep in hours • physical_activity, Physical activity level • stress_level, Stress level • sugar_consumption, Daily sugar intake level • alcohol_intake, Alcohol consumption in numeric score • salt_intake, Dietary salt intake in numeric score
Clinical	• bmi, BMI (kg/m2) • bmi_level, BMI category (WHO standard) • hbA1c_level, Long-term glucose indication in percentage

	• glucose, Blood glucose level • cholesterol, Total cholesterol level • triglycerides, Triglyceride level • blood_pressure, Blood pressure (mmHg) • crp_level, CRP level • homocysteine_level, Homocysteine level • systolic_bp, Systolic BP • diastolic_bp, Diastolic BP • heart_rate, Resting heart rate (bpm) • hdl, HDL cholesterol level • ldl, LDL cholesterol level
Disease Indicators	• diabetes, Whether the instance has diabetes or not • hypertension, Hypertension indicator • heart_disease, Heart disease indicator, how common heart disease is for a patient with a similar record. • family_history, Family history of chronic disease • low_hdl_cholesterol, Low HDL indicator • high_ldl_cholesterol, High LDL indicator • high_blood_pressure, High BP indicator
Others	• composite_key, Identifier that represents a demographic profile across (age group, gender, BMI category, smoking status, diabetes status combination) • source_dataset, Original source dataset • disease_flags, Disease combination code • sublabel, Classifies the exact kind of disease a person has • label, Binary target, classifies if a person has a disease or not.

2.2 Initial Data Quality Inspection

The raw dataset was inspected for missing values, duplicate records, unique values, categorical inconsistencies, numerical ranges, and unusual values before any cleaning was performed.

2.2.1 Missing Values

```
# Missing values
missing_values = df_original.isnull().sum()
missing_values[missing_values > 0]
```

✓ 0.4s

Series([], dtype: int64)

This screenshot shows that 0 values are missing in the 280985 instances in the dataset. It is highly unusual for a dataset of this size to be completely filled with values, and from the dataset, the author of the dataset justified that this dataset is based on 3 other datasets; the author may have removed or addressed instances during the integration process of creating this dataset.

2.2.2 Duplication Check

```
# Duplicate rows
print("Duplicate rows:", df_original.duplicated().sum())
```

✓ 1.7s

Duplicate rows: 2284

In the image above, the code shows that there are 2284 rows of duplicate entries in the data. For now, we are only identifying the duplicated rows, these rows will be revisited later on during the data cleansing stage to decide if they should be removed.

2.2.3 Unique Values Check

```
✓  
# Unique values in each column  
for column in df_original.columns:  
    print(f"{column}: {df_original[column].nunique()} unique values")  
2] ✓ 0.8s
```

The number of unique values were analysed for each column in order to recognise unexpected categories and unusual variable behaviour. Most of the categorical variables had small and consistent values. For example, gender contains 2 categories, and smoking, physical_activity, and stress_level each contain 3 categories. There were no apparent spelling errors, capitalisation, and formatting inconsistencies of category values.

The column 'composite_key' had a fairly low 192 unique values (for the total data set of 280,985 records). But this isn't too surprising because the column is a combination of characteristics instead of a unique identifier for each person.

The most unexpected discovery was with the 'heart_disease' column. The column seems like it would be a binary identifier representing whether or not an individual has heart disease. However it consists of 125 unique numerical values, including fractional values such as 0.08, 0.222222 and 0.032258. This shows that this column cannot be interpreted as a simple yes or no as we originally thought, and would require further interpretation in section 2.3.

```
composite_key: 192 unique values  
age_level: 4 unique values  
gender: 2 unique values  
bmi_level: 4 unique values  
smoking: 3 unique values  
diabetes: 2 unique values  
age: 88 unique values  
age_normalized: 151 unique values  
bmi: 14122 unique values  
hypertension: 137 unique values  
heart_disease: 125 unique values  
HbA1c_level: 192 unique values  
glucose: 154 unique values  
cholesterol: 323 unique values  
sleep_hours: 10092 unique values  
triglycerides: 525 unique values  
physical_activity: 3 unique values  
family_history: 2 unique values  
stress_level: 3 unique values  
low_hdl_cholesterol: 2 unique values  
high_ldl_cholesterol: 2 unique values  
blood_pressure: 197 unique values  
high_blood_pressure: 2 unique values  
sugar_consumption: 3 unique values  
crp_level: 9992 unique values  
...  
source_dataset: 3 unique values  
disease_flags: 8 unique values  
sublabel: 8 unique values  
label: 2 unique values  
Output is truncated. View as a scrollable element
```

2.2.4 Range Values

```
# Numerical ranges and summary statistics
df_original.describe().T
```

✓ 0.6s

	count	mean	std	min	25%	50%	75%	max
age	280985.0	49.665865	21.682350	2.000000	32.000000	50.000000	67.000000	89.000000
age_normalized	280985.0	0.547883	0.249222	0.000000	0.344827	0.551724	0.747126	1.000000
bmi	280985.0	27.490663	7.040373	10.010000	22.100000	27.320000	32.700000	95.690000
hypertension	280985.0	0.128218	0.187346	0.000000	0.000000	0.071429	0.217063	1.000000
heart_disease	280985.0	0.073390	0.138243	0.000000	0.000000	0.007299	0.115053	1.000000
HbA1c_level	280985.0	5.947475	0.954072	3.500000	5.384805	5.800000	6.865766	9.000000
glucose	280985.0	135.258646	38.477865	70.000000	100.000000	138.000000	160.000000	300.000000
cholesterol	280985.0	224.045525	35.525530	150.000000	202.000000	225.000000	243.000000	300.000000
sleep_hours	280985.0	6.954652	1.426412	4.000000	6.200000	6.975185	7.700000	10.000000
triglycerides	280985.0	178.995744	67.336085	50.000000	129.000000	179.000000	242.000000	400.000000
blood_pressure	280985.0	149.887200	4.148644	120.000000	148.946429	149.713320	150.720942	180.000000
crp_level	280985.0	7.489127	1.013898	0.003647	7.294863	7.458483	7.758562	14.997087
homocysteine_level	280985.0	12.465326	0.983375	5.000236	12.287763	12.488407	12.686175	19.999037
systolic_bp	280985.0	134.405105	20.542140	90.000000	126.000000	134.167413	143.000000	179.000000
diastolic_bp	280985.0	89.476720	13.667219	60.000000	84.000000	89.504587	95.000000	119.000000
alcohol_intake	280985.0	15.003336	6.830255	0.000000	12.000000	14.982818	17.900000	30.000000
salt_intake	280985.0	8.480577	2.960646	2.000000	7.200000	8.477600	9.800000	15.000000
heart_rate	280985.0	74.497343	11.402996	50.000000	70.000000	74.487247	79.000000	99.000000
hdl	280985.0	64.609147	15.938525	30.000000	58.000000	64.650655	72.000000	99.000000
ldl	280985.0	129.725919	27.352521	70.000000	118.000000	129.983051	141.000000	189.000000

The numerical values were analysed to show their maximum, median, average, and quartile values. Most column ranges shown above show plausible data. The columns cholesterol, glucose, hdl, ldl, triglycerides, and sleep hours all seemed to be reasonable. However there are some values which are unusual. One being a minimum value of 2, and the other is a maximum BMI of 95.69. These unusual values still cannot be considered errors from just their value, and we will need to analyse them more during data cleansing before any editing is done.

2.3 Data Interpretation and Assumptions

The initial inspection of the data revealed some important issues with how heart disease is represented in the data. The **heart_disease** column is represented as a range of values anywhere between 0 and 1, it is not a binary classification of true or false. The main goal of this project was to determine whether or not a person had a heart disease. This means that the **heart_disease** column cannot be used as the binary target.

This dataset also has a '**label**' column, this classifies each person under either 'normal' or 'abnormal', which is exactly the binary type we are looking for. However, 'abnormal' is way too vague and only identifies if a record is associated with a disease, it does not identify an exact disease.

The sublabel column provides a more precise classification for each record by assigning each record into the eight possible groupings of diseases; **N** for no listed disease, **DI** for diabetes, **HT** for heart disease, **HY** for hypertension, **DI-HT** for diabetes and heart disease, **DI-HY** for diabetes and hypertension, **HT-HY** for heart disease and hypertension, and **DI-HT-HY** for all three diseases. From these classifications, any sublabel that contains '**HT**' can be treated as a person having a heart disease. This is exactly the binary target we are looking for in this project, so the '**sublabel**' column will be used to construct a new binary heart disease target.

There are also several variables which represent similar data differently, for example **age**, **age_level**, and **age_normalized** all represent a person's age while **bmi** and **bmi_level** both represent the person's BMI. Later analysis will be carried out on these factors when deciding whether they are suitable to remain when modelling.

Disease related columns such as **sublabel**, **disease_flags**, **label** and others could directly reveal if there is a heart disease present. This means that they need to be removed from the predictor variables appropriately to prevent **data leakage**.

Finally, outlier values will not be automatically removed just because they are extreme. Instead they will first need to be investigated further during cleansing and analysis before deciding whether it should be kept, transformed or removed.

3. Data Cleansing and Transformation

3.1 Data Cleaning and Preparation

All cleaning and transformations were performed on a separate working copy of the original dataset, to ensure the original data remains untouched.

As noted in the initial data inspection, the current heart_disease column cannot serve as a binary target since it consists of a range of numerical values, rather than only two binary values. Therefore, **a new binary target variable was created** using the sublabel column. If a record had a sublabel of 'HT', 'DIHT', 'HTHY', or 'DIHTHY', it would be classified as having heart disease, while sublabels 'N', 'DI', 'HY', or 'DI_HY' would be classified as non-heart disease. This mapping was then verified across all eight sublabel values, confirming that all records were assigned to the intended category.

```
Duplicate rows:
0

Heart disease target distribution:
heart_disease_target
False    272810
True      5891
Name: count, dtype: int64
```

```
Target mapping:
heart_disease_target  False  True
sublabel
DI                    31903    0
DI_HT                 0    1398
DI_HT_HY              0    845
DI_HY                66472    0
HT                     0   2583
HT_HY                 0   1065
HY                   69787    0
N                   104648    0
```

```
# Apply the cleaning function
df_clean = clean_data(df_original)

print("Original shape:", df_original.shape)
print("Cleaned shape:", df_clean.shape)
print("Rows removed:", len(df_original) - len(df_clean))
```

✓ 0.8s

```
Original shape: (280985, 39)
Cleaned shape: (278701, 40)
Rows removed: 2284
```

The dataset had 2,284 duplicate records that were identified during the initial inspection and were removed from the dataset during cleaning. This reduced the dataset from 280,985 records to 278,701 records. A validation test showed that there were no duplicate records that remained.

No missing values were identified previously in the dataset, so it wasn't addressed during cleaning. The initial inspection also did not have apparent formatting and capitalization inconsistencies. All the identified unusual values have not been modified at this stage and will be further analysed in section 3.2.

Some disease related variables were also checked against the new heart disease target to discover predictors who **may reveal data leakage during later modeling**. The 'sublabel' essentially is our new target. The variable 'disease_flags' described the same disease combinations but written in another form. The 'label' variable did not uniquely classify heart diseases, however all records with heart disease were classified as 'Abnormal', which means that the column still contained information about the disease. The already existing 'heart_disease' variable also showed a relationship to the new target, but it was a range of values between 1 - 0. Other disease indicators, like 'diabetes' and 'hypertension' showed up in both heart disease cases and they did not directly tell us anything about the disease itself.

These variables have been kept in for phase 1, but it should be noted that the disease related variables that directly contain or reveal indicators of the heart disease classification will not be used in later modeling, because this would cause data leakage.

In this stage, invalid data types and formatting issues have been considered as well. All data types are stored correctly, with numerical variables as integers and floats, categorical variables as strings, and lastly output variables as booleans. Therefore, no further changes would need to be performed to convert data to their appropriate types.

Additional checks have been performed for invalid data types and formatting issues. The cleaned data showed that the numerical variables are stored correctly as integers and float types, all categorical variables are string type and the output variable is stored as Boolean. This means that we do not need to do any type conversions during cleaning. Categorical formatting was also consistent during the initial inspection so no modifications needed to be made.

The dataset contained no missing values therefore no action was taken. The unusual values that inspected in section 2.2.4 were analysed to determine if they should have been removed; during investigation all cases with the lowest reported value of age (2) had correctly been placed within the

'Young' group of ages, and the entry with the highest value in BMI (95.69) had also been correctly placed in the 'Obese' category. This means that these extreme values should be treated as valid in this stage, and will be for later outlier analysis in section 3.2.

3.2 Outlier Investigation and Treatment

The possibility of any numerical outliers in the dataset was analysed using the Interquartile Range (IQR) method of identifying outlier data. Values that were more than 1.5 times the IQR (*less than the first quartile or greater than the third quartile*) were flagged as possible outliers. These identified values were not automatically removed because a statistical outlier does not imply that data is incorrect.

Data on BMI was investigated first because the initial inspection of the raw data showed an unusual maximal BMI of 95.69. For BMI; The first quartile is 22.01 and third quartile is 32.70, with an resulting IQR of 10.69. This gave an upper boundary of 48.74 and a lower boundary of 5.98.

```
Number of BMI outliers: 1005
Q1: 22.01
Q3: 32.7
IQR: 10.690000000000001
Lower boundary: 5.974999999999998
Upper boundary: 48.73500000000001
```

Our dataset has a minimum BMI of 10.01 but this remains above the lower limit of 5.98 and so no lower range outliers were observed. However, there are 1,005 records with a BMI level that is above the upper boundary of 48.74, and were therefore flagged as potential statistical outlier

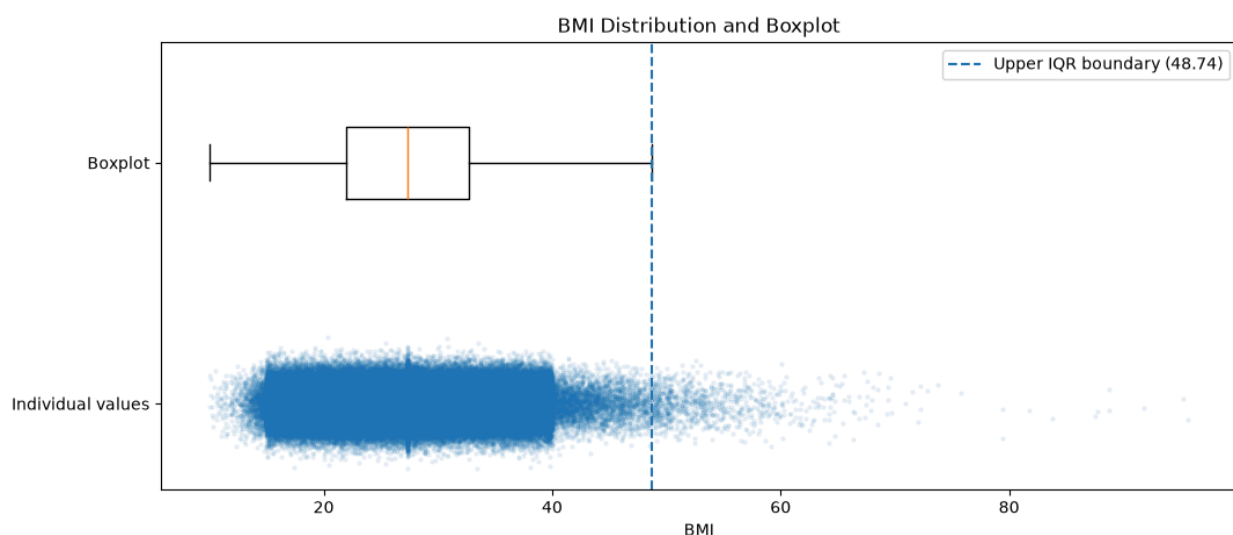


Figure 1.1, BMI Distribution and Boxplot created using matplotlib

The distribution shows that the flagged BMI values continue gradually past the upper IQR boundary instead of only appearing as few scattered and isolated numbers of values. These extreme cases were also compared with the already existing 'bmi_level' column, all of which showed the correct 'obese' classification. These observations show that there is no evidence to say that the 'outliers' are invalid. Because of this, the BMI values flagged by the IQR method were not removed or transformed. They will stay in the cleaned dataset as valid values.

3.3 Cleaned Dataset and Validation

The dataset was checked after cleaning in order to make sure any changes were applied correctly and that no issues were introduced in the process. Before cleaning, the dataset contained 280,985 rows and 39 columns. During the cleaning process, a duplicate check was run, and 2,284 duplicate rows were removed. Then, missing columns were tested in the dataset; however, none were found. A new column named heart_disease_target was added during this process. After the cleaning process and several checks, the new dataset contains 278,701 rows and 40 columns

The heart_disease_target column was checked against the sublabel column to make sure that the mappings were correct. Any sublabel values: HT, DI_HT, HT_HY and DI_HT_HY were all mapped to True, while everything else was mapped to False. There were no collisions in mapping.

After creating the new column, the numerical ranges before and after were compared. There were no changes in the minimum value for any numerical columns. For example, the BMI minimum and maximum values remained unchanged from 10.01 to 95.69, glucose was between 70 and 300, and the systolic blood pressure was between 90 and 179. This proved that removing all duplicate rows did not remove any unique minimum or maximum values.

All data types were checked again after cleaning. All previous numerical and categorical columns remained the same data type, while the new heart_disease_target column stores boolean values. Overall, multiple validation checks were run to confirm all cleaning changes were applied correctly, resulting in no unexpected issues. All duplicate rows were removed, there were no missing values found, the new column was created properly, and the original ranges from before cleaning were preserved.

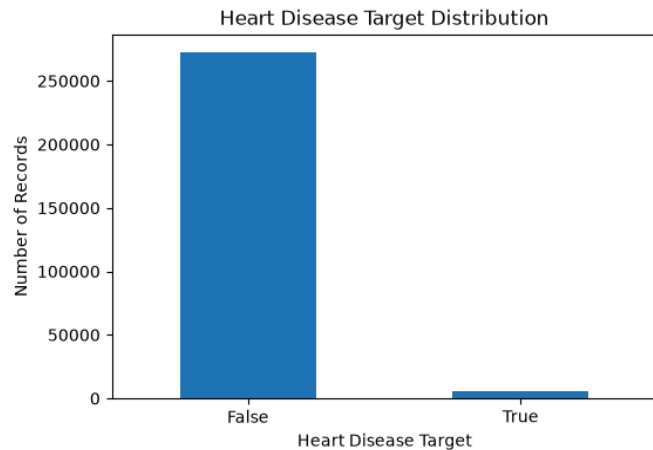
4. Exploratory Data Analysis

4.1 Descriptive and Distribution Analysis

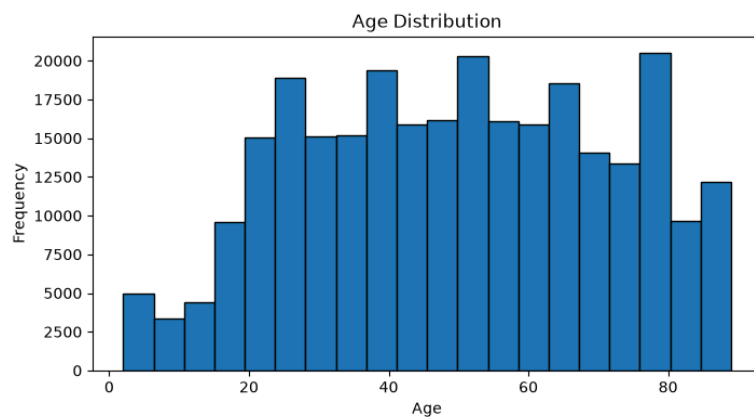
Analyse the overall characteristics of the important variables. Include relevant:

the target is very imbalanced: only 5,891 of 278,701 cleaned records are heart-disease cases

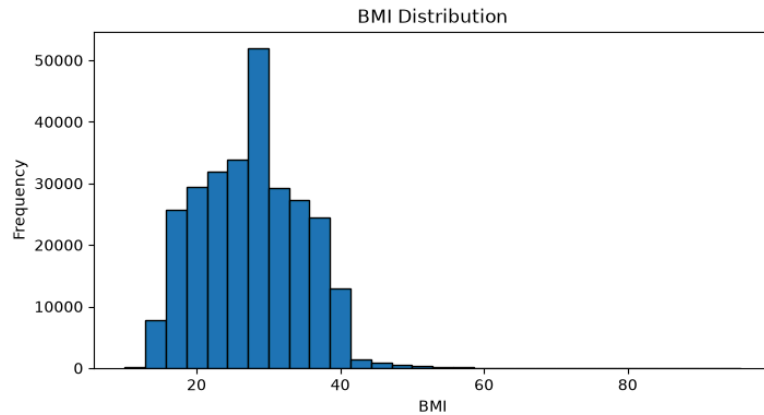
- ☐ descriptive statistics
- ☐ numerical distributions
- ☐ categorical frequencies
- ☐ histograms
- ☐ boxplots
- ☐ bar charts
- ☐ target/class distribution
- ☐ class imbalance



The bar graph of the distribution of heart disease between false and true shows a clear dominance in the number of instances with a heart disease target of false, with above 250,000 instances. In comparison, instances that have heart disease are only around 5,891 instances. Therefore, this dataset is heavily biased towards the dominant class (False), which means that the model trained on this data would likely predict that most cases have no heart disease. This satisfies the model accuracy, but will fail in trying to indicate whether an individual actually has heart disease.



This histogram shows a wide age distribution in the cleaned dataset. It ranges from age 2 to around age 89. The graph shows a multimodal shape, having multiple peaks in the age range of 20 to 80, which means there is a clear concentration of adults. In this dataset, there is a clear indication that the main focus of the age demographic is on adults, which makes perfect sense as adults make up the majority of heart disease cases.

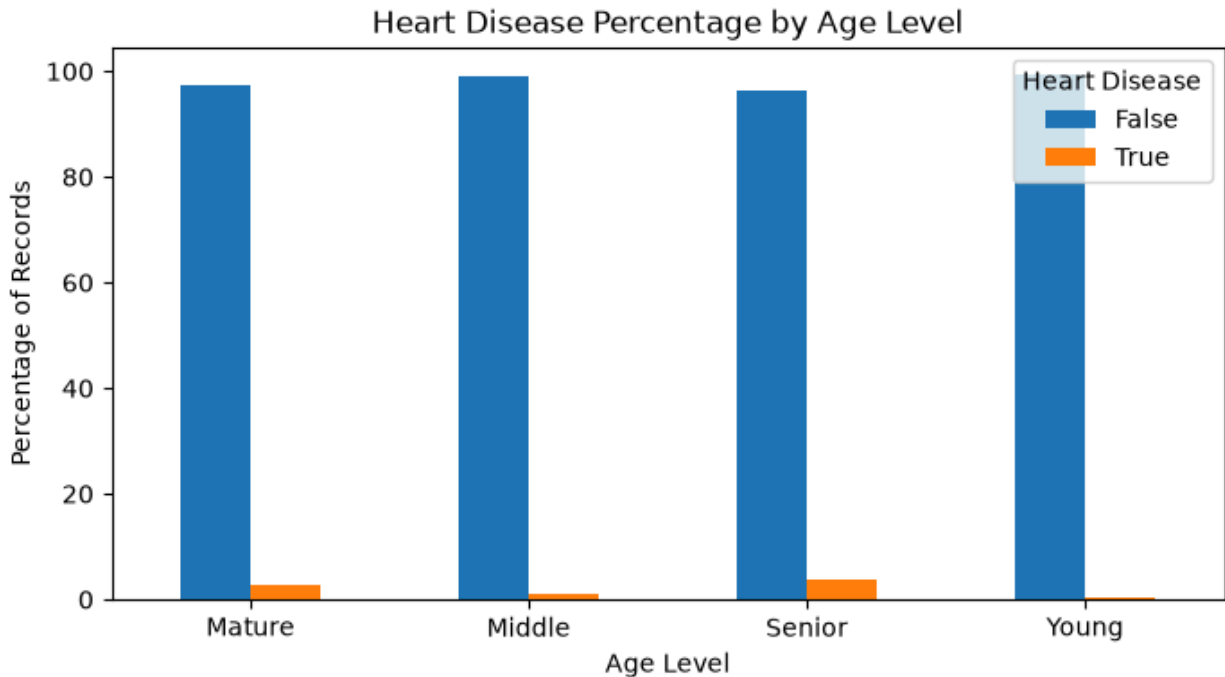


In this histogram of BMI distribution, the BMI ranges from 10 to 95. It shows a clear concentration at around 30 BMI with around 50000 instances. While the main bulk of the data appears to be a normal distribution shape, the graph is actually right-skewed, as the long range of BMI on the x-axis confirms that the range extends towards 80. The 95.69 BMI instance is validated to be an accurate instance from the cleaning process in section 3.1.

4.2 Relationships and Patterns in the Data

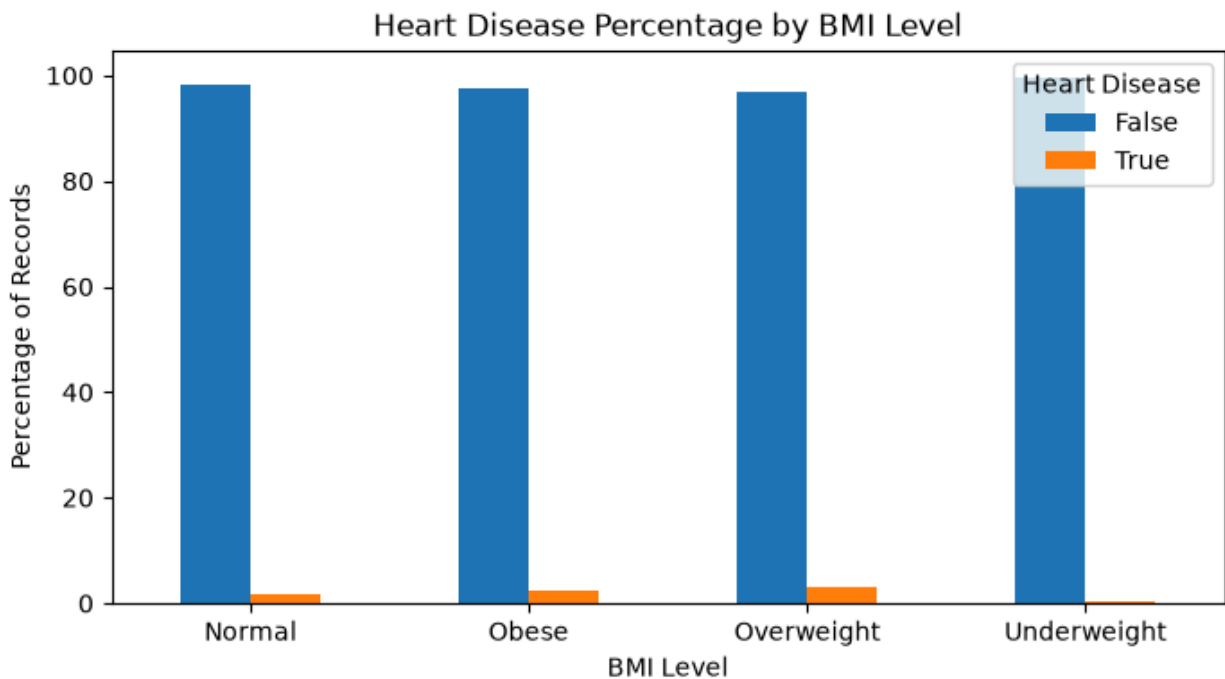
Investigate relationships between variables and with the target. Include relevant:

- ☐ scatter plots
- ☐ grouped comparisons
- ☐ correlation analysis
- ☐ correlation heatmap
- ☐ relationships with the target
- ☐ missingness patterns
- ☐ unusual patterns or clusters

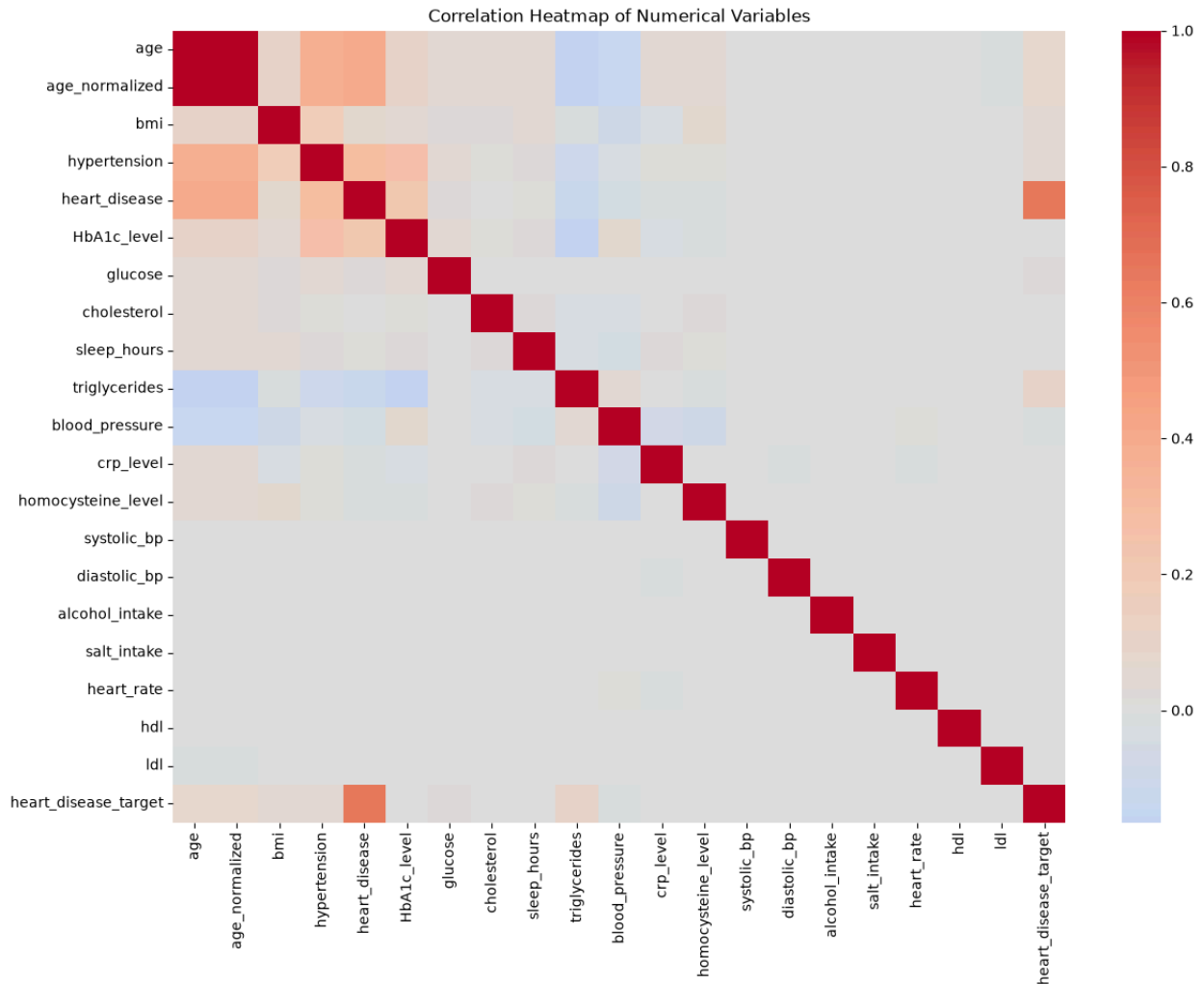


The records classified with heart disease in each group change as the age group gets higher. The young group has the lowest percentage of records with heart disease. The middle age group also has a very low percentage, but is still higher than the young age group. The mature age group has a higher percentage and starts to be more visible on the graph. The senior age group has the highest percentage of records classified with heart disease in this graph. This creates an increasing pattern of records with heart disease as the age group increases.

This relationship is consistent with age being a main characteristic for target distribution. We cannot state that age alone explains the presence of heart disease, due to most records in all age groups not stating heart disease. The cause of this is the strongly imbalanced dataset, with only 5,891 of 278,701 cleaned records. Out of these cleaned records, only 2.1% of them are classified as having heart disease. The age groups in this graph are very broad, so differences between the records in each group are hidden. Any patterns in the data depend on how boundaries are defined for each age group, so if it is tweaked, the graph will show different statistics.



When comparing the two variables, BMI categories against people with and without heart disease, for all BMI levels (normal, obese, overweight, and underweight), we can see that the percentage of people with heart disease is relatively small. The overweight and obese BMI categories show a greater percentage than the normal BMI category, while the underweight category has the lowest percentage of people with heart disease. Even though there is a difference in percentage across the different BMI categories, it does not seem that, from the graph, there is a strong indication of a relationship between the two variables. This relates to the heart disease bar graph because the low number of instances of heart disease is heavily unbalanced (fewer instances for people with heart disease), thus making the proportion of people with heart disease seem way lower. Although the percentages are small across all categories, there is still an indication of which category has the highest proportion of individuals who have heart disease, though not a strong link between them.



The heatmap above shows that most variable pairs have a weak linear correlation. This implies that the dataset covers different types of information like demographics, lifestyle, and health, instead of repeating data in each column. The reason why there are not many visible linear relationships is that there may be relationships between variables, but not linear ones.

There is a very clear linear relationship between age and age_normalised, with a correlation value of 1.0. However, this is due to age_normalised using only age, and making it more broad (numerical to categorical). These two variables should not be used as predictors for this exact reason.

Age also correlates with the heart_disease variable and moderately with the hypertension variable, with a correlation of 0.6. The new heart_disease_target variable is very strongly correlated with the existing heart_disease variable. Target leakage is introduced because the heart_disease variable

contains information about the result we are looking for. This heatmap provides evidence of why the heart_disease variable should be excluded as a predictor.

4.3 Key Findings and Dataset Implications

Bring the EDA together rather than ending with a collection of graphs. Summarise the most important findings about:

- ☐ dataset structure
- ☐ data quality
- ☐ important relationships
- ☐ imbalance
- ☐ unusual patterns
- ☐ apparent usefulness of the dataset for the intended problem
- ☐ issues that may need to be addressed later during modelling

5. Data Engineering Pipeline and Phase I Summary

5.1 Data Engineering Pipeline

Present the required pipeline diagram and briefly explain the overall process:

Raw Dataset → Acquisition → Inspection → Cleaning → Transformation → Validation → Cleaned Dataset → Exploratory Data Analysis

5.2 Phase I Summary and Next Steps

Briefly summarise what was achieved during Phase I and identify what the project will move into during Phase II, such as feature engineering and machine learning.

Heart Disease Target Note

- Heart disease is a broad category, not one specific condition. It can include coronary artery disease, heart failure, arrhythmias, valve disease, and cardiomyopathy.
- The dataset does not identify the specific type of heart disease.
- Therefore, the project should ask: *Can clinical and lifestyle measurements be used to predict whether a patient has heart disease?*
- Limitation: different heart conditions are grouped into one binary target.

References

1. Cardiovascular diseases are the leading cause of death globally (World Health Organisation, 2025).

2. Telfar Barnard, L. (2025, July). Heart disease in Aotearoa: Morbidity, mortality, and service delivery. Prepared for Kia Manawanui Trust, University of Otago.